

MATH 544: Real Analysis Qualifying Exam Prep

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Part I

Notes

These are a brief summary of Dr. Monica Torres' Fall 2025 MA 544 class notes. The notes are originally based on the book [2], but most have been taken directly from Dr. Torres' handwritten notes found in [1]. Unless specified otherwise, suppose X is a general set, $\mathcal{M}$ is a σ -algebra, μ is a generic measure, and λ is the Lebesgue measure.

1 Measure Theory

1.1 Set and Measure Theory

Definition 1.1.1: Outer Measure.

A function φ defined for every subset A of an arbitrary set X is called an **outer measure** on X if the following hold:

- (i) $\varphi(\emptyset) = 0$
- (ii) $0 \leq \varphi(A) \leq \infty$ for all $A \subset X$.
- (iii) $\varphi(A_1) \leq \varphi(A_2)$
- (iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for any countable collection $\{A_i\} \in X$.

Condition (iii) says that every outer measure is monotone, and (iv) says that φ is countably subadditive.

Definition 1.1.2.

Let φ be an outer measure on a set X . A set $E \subset X$ is called φ -measurable if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A \setminus E) \quad (1.1)$$

for all $A \subset X$.

Note that we have

$$\varphi(A) = \varphi[(A \cap E) \cup (A \cap E)^c] \leq \varphi(A \cap E) + \varphi(A \cap E^c) \quad (1.2)$$

So then to check that the set E is φ -measurable, we only need to check if

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A \setminus E) \quad (1.3)$$

Lemma 1.1.3.

A set $E \subset X$ is φ -measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q) \quad (1.4)$$

for any sets P, Q with $P \subset E$ and $Q \subset E^c$.

Proof. ($\Leftarrow$) Let $E \subset X$ and $A \subset X$. Define the following:

$$P := A \cap E \subset E \quad (1.5)$$

$$Q := A \setminus E \subset E^c \quad (1.6)$$

So then we have

$$\varphi(A) = \varphi(P \cup Q) = \varphi(P) + \varphi(Q) \quad (1.7)$$

$$= \varphi(A \cap E) + \varphi(A \setminus E) \quad (1.8)$$

Thus E is measurable.

($\Rightarrow$) Suppose E is measurable. Let $P \subset E$ and $Q \subset \tilde{E}$. Then

$$\varphi(P \cup Q) = \varphi[(P \cup Q) \cap E] + \varphi[(P \cup Q) \setminus E] \quad (1.9)$$

$$= \varphi((P \cap E) \cup (Q \cap E)) + \varphi((P \cap E^c) \cup (E^c \cap Q)) \quad (1.10)$$

$$= \varphi(P \cup \emptyset) + \varphi(\emptyset \cup Q) \quad (1.11)$$

$$= \varphi(P) + \varphi(Q) \quad (1.12)$$

□

Example 1.1

Let $x_0 \in X$ define

$$\varphi(A) = \begin{cases} 0 & x_0 \notin A \\ 1 & x_0 \in A \end{cases} \quad (1.13)$$

Here, φ is the Dirac measure concentrated at x_0 . φ is indeed an outer measure.

Definition 1.1.4: Equivalence.

Let A and B are said to be **equivalent** if there exists some $f : A \rightarrow B$ which is 1-to-1 and onto. We then say that $A \sim B$.

$\sim$ is an equivalence relation on a set X , so it is reflexive, symmetric, and transitive. We say that two sets are in the the same equivalence class if they have the same cardinality, defined by:

Definition 1.1.5: Cardinality.

Define the **cardinality** of a set $X = \{x_1, x_2, \dots, x_n\}$ by

$$|X| = n \quad (1.14)$$

We define the cardinality of common sets as follows:

$$|\mathbb{N}| = N_0 \quad (1.15)$$

$$|\mathbb{R}| = c \quad (1.16)$$

We say that a set X is a finite set if $|X| = n$ for $n < \infty$. if a set is not **finite**, then we say that it is **infinite**. If $X \sim \mathbb{N}$, then X is said to be **denumerable**. If $|X| = n$ or $|X| = \mathbb{N}$, then X is **countable**. Otherwise, it is **uncountable**. We say that X is **at most countable** if X is either finite or countable.

Definition 1.1.6.

Let $\alpha = |A|$ and $\beta = |B|$. We say that $\alpha \leq \beta$ if and only if there is some subset $B_1 \subset B$ such that $A \sim B_1$. We say that $\alpha < \beta$ if there does not exist a set $A_1 \subset A$ such that $A_1 \sim B$

Theorem 1.1.7.

If $|A| \leq |B|$ and $|B| \leq |A|$, then we have that $|A| = |B|$.

Definition 1.1.8.

Let F be the family of all functions $f : B \rightarrow A$. Then we have

$$|A| + |B| = |A \cup B| \quad (1.17)$$

$$A \cap B = \emptyset \quad (1.18)$$

$$|A|^{|B|} = |F| \quad (1.19)$$

By Definition 1.1.8, we also have the following:

$$N_0 + N_0 = N_0 \quad (1.20)$$

$$c + c = c \quad (1.21)$$

$$2^{N_0} = c \quad (1.22)$$

$$2^\alpha > \alpha \quad (1.23)$$

Theorem 1.1.9.

Let φ be an outer measure on an arbitrary set X . Then

- (i) If $\varphi(E) = 0$, then E is φ -measurable.
- (ii) $\emptyset$ and X are φ -measurable.
- (iii) If E_1, E_2 are φ -measurable, then $E_1 \setminus E_2$ is φ -measurable.
- (iv) Let $\{E_i\}$ be a countable collection of disjoint φ -measurable sets, then $\bigcup_{i=1}^{\infty} E_i$ is φ -measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i) \quad (1.24)$$

More generally, we have that if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \quad (1.25)$$

where $S = \bigcup_{i=1}^{\infty} E_i$.

Proposition 1.1.10.

If E is φ -measurable, then E^c is φ -measurable.

Proof. Since both E and X are φ -measurable, then $X \setminus E$ is measurable, and we know that $X \setminus E = E^c$ so we are done. $\square$

Lemma 1.1.11.

Let $\{E_j\}$ be a sequence of arbitrary sets. Then, there exists a sequence of disjoint sets $\{A_i\}$ such that $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i \quad (1.26)$$

Furthermore if each E_i is φ -measurable

Theorem 1.1.12.

If $\{E_i\}$ is a sequence of φ -measurable sets in X , then $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$ are φ -measurable.

Proof. We know from Lemma 1.1.11 that if we define A_i to be disjoint measurable sets such that

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i \quad (1.27)$$

then $\bigcup_{i=1}^{\infty} A_i$ is φ -measurable and hence $\bigcup_{i=1}^{\infty} E_i$ is φ -measurable. For the intersection we have by De Morgan's Laws,

$$\bigcap_{i=1}^{\infty} E_i = \left(\bigcup_{i=1}^{\infty} E_i^c \right)^c \quad (1.28)$$

which is a countable intersection, and therefore also φ -measurable. $\square$

Definition 1.1.13: σ -Algebra.

We define a σ -algebra on X to be a nonempty collection Σ of sets $E \subset X$ such that

- (i) If $E \in \Sigma$ then $E^c \in \Sigma$.
- (ii) For all $E_i \in \Sigma$, we have $\bigcup_{i=1}^{\infty} E_i \in \Sigma$

Proposition 1.1.14.

If φ is an outer measure on X , then the class of φ -measurable sets form a σ -algebra.

Corollary 1.1.15.

Let φ be an outer measure on X and $\{E_i\}$ be a countable collection of φ -measurable sets. Then

(i) If $E_1 \subset E_2$, and $\varphi(E_1) < \infty$ then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1) \quad (1.29)$$

(ii) If $E_i \subset E_{i+1}$ then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i) \quad (1.30)$$

(iii) If $E_i \supset E_{i+1}$ and $\varphi(E_{i_0}) < \infty$ for some i_0 , then

$$\varphi\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i) \quad (1.31)$$

1.2 Borel Sets

We now look to introduce the concept of Borel sets that we will incorporate into our existing measure theory. In a general topological space, we consider the elements of the smallest σ -algebra which contains all open sets to be **Borel sets**. We have the following definitions to help define the Borel σ -algebra:

Definition 1.2.1: Regular Outer Measure.

An outer measure φ is considered **regular** if for each $A \subset X$ there exists a φ -measurable set B , $A \subset B$ such that $\varphi(A) = \varphi(B)$.

Definition 1.2.2: Borel Regular Outer Measure.

A regular outer measure φ is **Borel** if the set B in Definition 1.2.1 is a Borel set.

Definition 1.2.3: Borel Outer Measure.

An outer measure φ on a topological space X is **Borel** if all Borel sets on X are φ -measurable.

We say that a Borel outer measure φ is **finite** if $\varphi(X) < \infty$.

Definition 1.2.4: Carathéodory Outer Measure.

A measure φ on a metric space (X, ρ) is called a **Carathéodory outer measure** if

$$\varphi(A \cup B) = \varphi(A) + \varphi(B) \quad (1.32)$$

for $A, B \subset X$ with $d(A, B) > 0$.

Recall, that in a metric space with metric ρ we have that

$$d(A, B) = \inf\{\rho(a, b) : a \in A, b \in B\} \quad (1.33)$$

Theorem 1.2.5.

If φ is a Carathéodory outer measure on a metric space X , then all closed sets are φ -measurable.

Proof. The set of all measurable sets $\mathcal{M}$ is a σ -algebra which contains all closed sets, and therefore all open sets as well as the intersection of closed sets is an open set. Since Borel sets are all open, then $\mathcal{M}$ contains all Borel sets. Since $\mathcal{M}$ is the set of all φ measurable sets, then all Borel sets are φ -measurable. $\square$

Theorem 1.2.6.

Suppose $\mathcal{B}$ is a family of subsets of a topological space X that contains all open and closed subsets of X . Suppose also that $\mathcal{B}$ is closed until all countable unions and countable intersections. Then $\mathcal{B}$ contains all Borel sets.

Corollary 1.2.7.

Let φ be a Carathéodory outer measure on a metric space X , then all Borel sets of X are φ -measurable.

Theorem 1.2.8.

Let φ be a Borel outer measure on X , and B be a Borel set. If $B \subset \bigcup_{i=1}^{\infty} V_i$ for V_i open sets with $\varphi(V_i) < \infty$, then for all $\varepsilon > 0$ there exists some $W \supset B$ with W open such that $\varphi(W \setminus B) < \varepsilon$.

1.3 Lebesgue Measure

We wish to define an Carathéodory outer measure on $\mathbb{R}^n$. To do this, consider the closed n -dimensional intervals

$$I = \{x : a_i \leq x_i \leq b_i, \quad i = 1, 2, \dots, n\} \quad (1.34)$$

and their volumes

$$v(I) = \prod_{i=1}^n (b_i - a_i) \quad (1.35)$$

So then we can redefine

$$I = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n] = I_1 \times I_2 \times \cdots \times I_n \quad (1.36)$$

Definition 1.3.1: Lebesgue Measure.

Let $E \subset \mathbb{R}^n$. Define

$$\lambda^*(E) = \inf \left\{ \sum_{I \in S} v(I) \right\} \quad (1.37)$$

where the infimum is taken over all countable collections S of closed intervals I such that $E \subset \bigcup_{I \in S} I$

Theorem 1.3.2.

For a closed interval $I \subset \mathbb{R}^n$, we have

$$\lambda^*(I) = v(I) \quad (1.38)$$

Theorem 1.3.3.

λ^* as defined in Definition 1.3.1 is a Carathéodory outer measure.

Theorem 1.3.4.

The following are equivalent for a Lebesgue measure:

- (i) $E \subset \mathbb{R}^n$ is λ^* -measurable.
- (ii) For every $\varepsilon > 0$ there exists an open U with $E \subset U$ such that $\lambda^*(U \setminus E) < \varepsilon$
- (iii) There is a G_δ set $U \supset E$ such that $\lambda^*(U \setminus E) = 0$
- (iv) For each $\varepsilon > 0$, there exists some closed $F \subset E$ such that $\lambda^*(E \setminus F) < \varepsilon$.
- (v) There is a F_σ set $F \subset E$ such that $\lambda^*(E \setminus F) = 0$.

1.4 The Cantor Set

Let $E_0 = [0, 1]$. Now, take $E_1 = E_0 \setminus (1/3, 2/3) = [0, 1/3] \cup [2/3, 1]$. Continue removing the middle third of each interval to get

$$E_2 = [0, 1/9] \cup [2/9, 1/3] \cup [6/9, 7/9] \cup [8/9, 1] \quad (1.39)$$

We can continue this to get E_n with the following properties:

- (a) $E_1 \supset E_2 \supset E_3 \supset \dots$
- (b) E_n is the union of 2^n intervals each of length $1/3^n$.

We then define

Definition 1.4.1: Cantor Set.

Let the cantor set be the intersection of all sets E_n , that is

$$C := \bigcap_{n=1}^{\infty} E_n \quad (1.40)$$

Clearly we have that $C \neq \emptyset$ since it contains $\{0, 1\}$. We also have that C is closed since it is the countable union of closed sets. It is clearly also bounded. Since it is closed and bounded, Heine-Borel says it is a compact set. Since each E_n is measurable, C is also measurable since it is the union of measurable sets. We can calculate the measure since clearly $\lambda^*(E_1) < \infty$. We see

$$\lambda(C) = \lambda\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} \lambda(E_n) = \lim_{n \rightarrow \infty} \frac{1}{3^n} 2^n = 0 \quad (1.41)$$

So then the Cantor set has measure zero.

Proposition 1.4.2.

C is nowhere dense in $[0, 1]$.

Proof. We note that since C is closed $\overline{C} = C$, and therefore $\lambda(\overline{C}) = \lambda(C) = 0$. Thus, $\overline{C}$ cannot contain any open interval, since this would force $\lambda(\overline{C}) > 0$. Since $\overline{C}$ cannot contain any closed sets, it must be no-where dense in $[0, 1]$. $\square$

Definition 1.4.3: Perfect Set.

Let X be a metric space. We say that $E \subset X$ is **perfect set** if E is closed and if every point of E is a limit point of E .

Example 1.2

- $[0, 1] \cup \{2\}$ is not a perfect set.
- $\mathbb{R}^2$ is a perfect set.
- Any closed interval $I \subset \mathbb{R}^n$ is a perfect set.

Theorem 1.4.4.

The Cantor set C is perfect.

Proof. We already know that C is closed, so all that is left is to show that every point in C is a limit point. Let $x \in C$ and let S be any open interval such that $x \in S$. Let I_n be the interval of E_n which contains x , since $x \in \bigcap_{n=1}^{\infty} E_n$. For some n large enough, we have that $I_n \subset S$. Since $x \in I_n$, we have that

$$(S \setminus \{x\}) \cap C \neq \emptyset \quad (1.42)$$

meaning that x is a limit point of C . Since x is arbitrary, we can conclude that C is indeed perfect. $\square$

Theorem 1.4.5.

Let P be a nonempty perfect set in $\mathbb{R}^n$. Then P is uncountable.

Proof. Since P has limit points, P must be an infinite set. For the sake of contradiction, suppose P is countable, so then

$$P = \{x_1, x_2, x_3, \dots\} \quad (1.43)$$

Consider $B_{r_1}(x_1) = \{y \in \mathbb{R}^n : |y - x_1| < R_1\} := V_1$. Since x_1 is a limit point, $B_{r_1}(x_1) \cap P$ contains an infinite number of points of P . We don't know where x_2 is, but clearly there exists $V_2 := B_{r_2}(z)$ for some $z \in P$ such that

- (i) $\overline{V}_2 \subset V_1$
- (ii) $x_1 \notin \overline{V}_2$
- (iii) $V_2 \cap P \neq \emptyset$

We wish to proceed via induction on open balls $\{V_n\}$ satisfying:

- (i) $\overline{V}_{n+1} \subset V_n$
- (ii) $x_n \notin \overline{V}_{n+1}$

(iii) $V_{n+1} \cap P \neq \emptyset$

We can define $K_n := \bar{V}_n \cap P$ which is compact since it is closed and bounded. Note that $x_n \notin K_{n+1}$ implies:

$$\bigcap_{n=1}^{\infty} K_n = \emptyset \quad (1.44)$$

We also know that $K_n \neq \emptyset$ and $K_1 \supset K_2 \supset K_3 \subset \dots$, so then

$$\bigcap_{n=1}^{\infty} K_n \neq \emptyset \quad (1.45)$$

Which contradicts (1.44), so then P is uncountable. $\square$

Corollary 1.4.6.

The Cantor set is uncountable.

This makes the Cantor set interesting, since C is both countable and of measure zero.

1.5 Nonmeasurable Sets

Theorem 1.5.1.

There exists a set $E \subset \mathbb{R}$ that is not Lebesgue Measurable.

Proof. Let $x, y \in \mathbb{R}$. We say that $x \sim y$ if $x - y \in \mathbb{Q}$ for some equivalent relation $\sim$. Define this equivalence relation that if $x \in \mathbb{R}$, then E_x is the equivalence class that contains x .

If $x \in \mathbb{Q}$, then $\mathbb{Q} = E_x$, and $E_x = \{x + r_i : r_i \in \mathbb{Q}\}$.

Each equivalence class is countable, and since $\mathbb{R}$ is uncountable, there must be an uncountable number of equivalence classes. By the Axiom of choice, we have that there must exist some S such that for each equivalence class E , $S \cap E$ consists precisely of one point.

If $x, y \in S$ then $x - y \notin \mathbb{Q}$, since otherwise they would be in the same equivalence class, which contradicts how we define S . We then define

$$D_s := \{x - y : x, y \in S\} \quad (1.46)$$

is a subset of the irrational number and therefore cannot contain any interval.

We have that since $\mathbb{R}$ is a countable union of rational sets, as in:

$$\mathbb{R} = \bigcup_{i=1}^{\infty} (S + r_i) \quad (1.47)$$

Since $\mathbb{Q}$ is countable and $S + r_i = \{x + r_i : x \in S\}$. Let $x \in \mathbb{R}$, so then $x \in E_x$, and $S \cap E_x = \{z\}$. If $z = x$, then $x \in S + 0$. If $z \neq x$, then $x - z = r_i$ for some $r_i \in \mathbb{Q}$, which we can then rearrange to get $x = z + r_i$, so then $x \in S + r_i$. We proceed by contradiction, and assume S is measurable. So then we have

$$\lambda(\mathbb{R}) = \lambda\left(\bigcup_{i=1}^{\infty} S + r_i\right) \leq \sum_{i=1}^{\infty} \lambda(S + r_i) \quad (1.48)$$

Since $\lambda(\mathbb{R}) > 0$, we have

$$\lambda(S) = \lambda(S + r_i) > 0 \quad (1.49)$$

Since otherwise $\lambda(\mathbb{R})$ would be 0. If $\lambda(S) < \infty$, then (1.46) yields a contradiction. If $\lambda(S) = \infty$, then $0 < \lambda(S \cap I_n) < \infty$ for every $I_n = [-n, n]$ and $S \cap I_n$ is measurable. Since $D_{S \cap I_n} \subset D_S$ does not contain any interval by (1.46), then we can apply the following Lemma 1.5.2 applied to $S \cap I_n$ gives contradiction. $\square$

Lemma 1.5.2.

If $S \subset \mathbb{R}$ is a Lebesgue measurable set of positive and finite measure, then the set of differences

$$D_S = \{x - y : x, y \in S\} \quad (1.50)$$

contains interval.

1.6 Lebesgue-Stieltjes Measure

Definition 1.6.1: Lebesgue-Stieltjes Measure.

Let λ_f^* be an outer measure defined on $\mathbb{R}$ by

$$\lambda_f^*(E) = \inf \left\{ \sum_{h_k \in \mathcal{F}} \alpha_f(h_k) \right\} \quad (1.51)$$

where the infimum is taken over all countable collections $\mathcal{F}$ of half-open intervals of the form $(a_k, b_k]$ such that $E \subset \bigcup_{h_k \in \mathcal{F}} h_k$ and $\alpha_f((a_k, b_k]) = f(b_k) - f(a_k)$.

Theorem 1.6.2.

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a nondecreasing function, then λ_f^* is a Carathéodory outer measure on $\mathbb{R}$.

Since it is an Carathéodory outer measure, it is then a Borel outer measure.

Theorem 1.6.3.

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is nondecreasing and right continuous, then

$$\lambda_f((a, b]) = f(b) - f(a) \quad (1.52)$$

This theorem means that there is an identification between Lebesgue-Stieltjes measures and non-decreasing right continuous functions.

Theorem 1.6.4.

Suppose μ is finite Borel outer measure on $\mathbb{R}$ and let

$$f(x) = \mu((-\infty, x]) \quad (1.53)$$

Then the Lebesgue-Stieltjes measure λ_f agrees with μ on all Borel sets.

Proof. Let μ be a finite Borel outer measure on $\mathbb{R}$. Define

$$f(x) = \mu((-\infty, x]) \quad (1.54)$$

Clearly, f is nondecreasing. We now fix $x_0 \in \mathbb{R}$ and let $\{x_j\}$ such that $x_j \rightarrow x_0$, $x_j > x_0$, and $x_{j+1} < x_j$. We have

$$\bigcap_{j=1}^{\infty} (-\infty, x_j] = (-\infty, x_0] \quad (1.55)$$

So then take some $x \in (-\infty, x_0]$. Clearly $x \in (-\infty, x_j]$ for all j , so then $x \in \bigcap_{j=1}^{\infty} (-\infty, x_j]$. Now we have if $x \notin (-\infty, x_j]$, then $x > x_0$. Then there exists some x_j such that $x_0 < x_j < x$. Then $x \notin (-\infty, x_j]$ and therefore $x \notin \bigcap_{j=1}^{\infty} (-\infty, x_j]$. From (1.55), we have

$$f(x_0) = \mu((-\infty, x_0]) = \mu \left(\bigcap_{j=1}^{\infty} (-\infty, x_j] \right) = \lim_{j \rightarrow \infty} \mu((-\infty, x_j]) = \lim_{j \rightarrow \infty} f(x_j) \quad (1.56)$$

So then $f(x_0) = \lim_{j \rightarrow \infty} f(x_j)$, and f is right continuous. Now, let λ_f be the Lebesgue-Stieltjes measure. Then $\mu \mapsto f \mapsto \lambda_f$. On the other hand, we have

$$\mu((a, b]) = \mu((-\infty, b] \setminus (-\infty, a]) = \mu((-\infty, b]) - \mu((-\infty, a]) = f(b) - f(a) \quad (1.57)$$

But by Theorem 1.6.3 gives us

$$\lambda_f((a, b]) = f(b) - f(a) \quad (1.58)$$

Therefore, μ and λ_f agree on all half open intervals. Since every open set in $\mathbb{R}$ is a countable union of disjoint half open intervals, then μ and λ_f agree on all open sets. Then by Theorem 1.2.8 we have that μ and λ_f agree on all Borel sets. $\square$

1.7 Hausdorff Measure

Definition 1.7.1: Hausdorff Measure.

Let $\varepsilon > 0$ and $E \subset \mathbb{R}^n$ and $0 \leq s \leq n$. Let

$$H_\varepsilon^s(E) = \inf \left\{ \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\text{diam } E_i)^s : E \subset \bigcup_{i=1}^{\infty} E_i, \text{diam } E_i < \varepsilon \right\} \quad (1.59)$$

Where

$$\alpha(s) = \frac{\pi^{s/2}}{\Gamma(\frac{s}{2} + 1)} \quad (1.60)$$

and

$$\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} dx, \quad 0 < t < \infty \quad (1.61)$$

Notice that $\varepsilon_1 < \varepsilon_2$ implies that $H_{\varepsilon_1}^s(E) \geq H_{\varepsilon_2}^s(E)$. We can extend our definition in (1.59) to s -dimensions by

$$H^s(E) := \lim_{\varepsilon \rightarrow 0} H_\varepsilon^s(E) = \sup_{\varepsilon > 0} H_\varepsilon^s(E) \quad (1.62)$$

We can also extend to any arbitrary metric space X where

$$\text{diam } E = \sup_{p, q \in E} \{d(p, q)\} \quad (1.63)$$

Recall that $\text{diam } \overline{E} = \text{diam } E$, so the sets E_i in the covering can be taken to be closed arbitrarily.

Proposition 1.7.2.

For each $0 \leq s \leq n$, H^s is a Carathéodory outer measure.

When $s \in \mathbb{N}$, our definition of $\alpha(s)$ is the Lebesgue measure of the unit ball in $\mathbb{R}^s$. We can further define when $s = 0$, since $\alpha(0)2^{-0}(\text{diam } E)^0 = 1$ for any set E . Therefore, H^0 is a counting measure on $\mathbb{R}^n$. We also have the following proposition about the relation of s and n :

Proposition 1.7.3.

$H^s \equiv 0$ for all $s > n$.

Definition 1.7.4: Hausdorff Dimension.

The **Hausdorff dimension** of an arbitrary set $A \subset \mathbb{R}^n$ is the number $0 < \delta_A \leq n$ such that

$$\delta_A := \inf \{t : H^t(A) = 0\} \quad (1.64)$$

This definition is equivalent to finding the unique number δ_A such that when $s < \delta_A$ then $H^s(A) = \infty$, and when $s > \delta_A$, then $H^s(A) = 0$.

Proposition 1.7.5.

If $s = \delta_A$ then exactly one of the following is true:

- (a) $H^s(A) = 0$
- (b) $H^s(A) = \infty$
- (c) $0 < H^s(A) < \infty$

The opposite direction is also true, since if $0 < H^s(A) < \infty$ then we have $\delta_A = s$.

Example 1.3

Any countable Set $A \subset \mathbb{R}^n$ has Hausdorff dimension of 0. We have that $H^0(A) = \infty$ and $H^t(A) = 0$, for any $t > 0$. The opposite; however, is not true. If the Hausdorff dimension of A is 0, it is not necessarily countable. For example, there always exists a cantor set $C(s) \subset [0, 1]$ such that $C(s)$ is uncountable and its Hausdorff dimension is zero.

Proposition 1.7.6.

$$\delta_{C(\gamma)} = \frac{\log 2}{\log(1/\gamma)} \quad (1.65)$$

For the basic $\gamma = 1/3$ cantor set we have

$$\delta_{C(\gamma)} = \frac{\log 2}{\log 3} \approx 0.63 \quad (1.66)$$

Proof.

$$H_{2\beta^k}^2(C(\gamma)) \leq \sum_{j=1}^{2^k} \alpha(s) 2^{-s} (\text{diam } I_{k,j})^s \quad (1.67)$$

$$= \sum_{j=1}^{2^k} \alpha(s) 2^{-s} (\beta^k)^s \quad (1.68)$$

$$= \alpha(s) 2^{-s} 2^k \beta^k \quad (1.69)$$

Pick some s such that $2\beta^s = 1$. Then we see

$$H^s(C(\gamma)) = \lim_{k \rightarrow \infty} H_{2\beta^k}^2(C(\gamma)) \leq \alpha(s) 2^{-s} < \infty \quad (1.70)$$

So then

$$0 < H^s(C(\gamma)) < \infty \quad (1.71)$$

implies

$$\delta_{C(\gamma)} = s = \frac{\log 2}{\log(1/\beta)} \quad (1.72)$$

□

1.8 Measure Spaces

Definition 1.8.1: Measure.

Let X be a set and $\mathcal{M}$ be a σ -algebra of subsets of X . Define a general **measure** $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that

(i) $\mu(\emptyset) = 0$.

(ii) If $\{E_i\}$ is a sequence of disjoint sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i) \quad (1.73)$$

Definition 1.8.2: Measure Space.

We say that $(X, \mathcal{M}, \mu)$ is called a **measure space** and the sets in $\mathcal{M}$ are called μ -**measurable sets**.

Definition 1.8.3: Borel Measure.

If $\mathcal{M}$ consists of the family of Borel sets in a metric space X , then μ is a **Borel measure**.

A measure is finite if $\mu(X) < \infty$ for all $X \in \mathcal{M}$.

Definition 1.8.4: σ -finite Measure.

A measure μ is σ -finite if

$$X = \bigcup_{i=1}^{\infty} E_i \quad (1.74)$$

for some $\{E_i\}$ with $\mu(E_i) < \infty$

Theorem 1.8.5: Properties of Measure Spaces.

Let $(X, \mathcal{M}, \mu)$ be a measure space. Suppose $\{E_i\}$ is a sequence of sets in $\mathcal{M}$. Then we have

(i) If $E_1 \subset E_2$, then $\mu(E_1) \leq \mu(E_2)$.

(ii) If $E_1 \subset E_2$, and $\mu(E_1) < \infty$, then

$$\mu(E_2 \setminus E_1) = \mu(E_2) - \mu(E_1) \quad (1.75)$$

(iii)

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i) \quad (1.76)$$

(iv) If $E_i \subset E_{i+1}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i) \quad (1.77)$$

(v) If $\mu(E_i) < \infty$ for some i , then

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i) \quad (1.78)$$

Definition 1.8.6: Complete Measure Space.

A measure μ with all subsets of sets of μ -measure zero are measurable, is **complete**. A space $(X, \mathcal{M}, \mu)$ with a complete μ is called a **complete measure space**.

Not every abstract measure space is complete, but the following theorem allows us to construct the completion.

Theorem 1.8.7.

Suppose $(X, \mathcal{M}, \mu)$ is a measure space. Define

$$\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subset B, B \in \mathcal{M}, \mu(B) = 0\} \quad (1.79)$$

and define $\overline{\mu}$ on $\overline{\mathcal{M}}$ by

$$\overline{\mu}(A \cup N) = \mu(A) \quad (1.80)$$

Then

(i) $\overline{\mathcal{M}}$ is a σ -algebra.

(ii) $\overline{\mu}$ is a complete measure on $\overline{\mathcal{M}}$ and $(X, \overline{\mathcal{M}}, \overline{\mu})$ is a complete measure space.

(iii) $\overline{\mu}$ is the only complete measure on $\overline{\mathcal{M}}$ that is an extension of μ .

Proof. (i) $\overline{\mathcal{M}}$ is closed under countable unions, since

$$\bigcup_{i=1}^{\infty} (A_i \cup N_i) = \left(\bigcup_{i=1}^{\infty} A_i\right) \cup \left(\bigcup_{i=1}^{\infty} N_i\right) \quad (1.81)$$

and $\bigcup_{i=1}^{\infty} N_i \subset \bigcup_{i=1}^{\infty} B_i$ with $\mu(\bigcup_{i=1}^{\infty} B_i) = 0$. Let $A \cup N \in \overline{\mathcal{M}}$ and $N \subset B$ with $\mu(B) = 0$. Without loss of generality, we can assume $A \cap N = \emptyset$ since otherwise we work with $A \cup (N \setminus A)$ and $N \setminus A \subset B \setminus A$ with $B \setminus A \in \mathcal{M}$, $\mu(B \setminus A) \leq \mu(B) = 0$. With A and N disjoint, we have $A \cup N = (A \cup B) \cap ((B^c \cup N) \cup (A \cup B))$. So then we see

$$(A \cup N)^c = (A \cup B)^c \cup ((B^c \cup N) \cup (A \cap B))^c \quad (1.82)$$

$$= (A \cup B)^c \cup ((B^c \cup N)^c \cap (A \cap B)^c) \quad (1.83)$$

$$= (A \cup B)^c \cup ((B \cap N^c) \cap (A \cap B)^c) \quad (1.84)$$

$$= (A \cup B)^c \cup ((B \setminus N) \setminus (A \cap B)) \quad (1.85)$$

Then since $(A \cup B)^c \in \mathcal{M}$ and $(B \setminus N) \setminus (A \cap B) \subset B$ with $\mu(B) = 0$, we have that $\overline{\mathcal{M}}$ is a σ -algebra.

(ii) Suppose $A_1 \cup N_1 = A_2 \cup N_2$ for $N_1 \subset B_1$, and $N_2 \subset B_2$ where $\mu(B_1) = \mu(B_2) = 0$. Then

$$\bar{\mu}(A_1 \cup N_1) = \mu(A_1) \leq \mu(A_2 \cup B_2) \leq \mu(A_2) + \mu(B_2) = \mu(A_2) = \bar{\mu}(A_2 \cup N_2) \quad (1.86)$$

So then $\bar{\mu}(A_1 \cup N_1) \leq \bar{\mu}(A_2 \cup N_2)$. The same calculation in reverse says the opposite, so we can conclude that $\bar{\mu}(A_1 \cup N_1) = \bar{\mu}(A_2 \cup N_2)$. So then $\bar{\mu}$ is complete.

(iii) It is left as an exercise to show the uniqueness of the completion. □

2 Measurable Functions

2.1 The Definition of a Measurable function

Definition 2.1.1: Measurable Function.

suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \rightarrow Y$ is called **measurable** with respect to $\mathcal{M}$ and $\mathcal{N}$ if $f^{-1}(E) \in \mathcal{M}$ for any $E \in \mathcal{N}$.

If Y is a topological space, then it is always assumed that $\mathcal{N}$ is the σ -algebra of Borel sets. In this case, we can denote $\mathcal{N}$ by $\mathcal{B}$ and f is measurable when $E \in \mathcal{B}$ is Borel.

Proposition 2.1.2.

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then E is measurable if and only if χ_E is measurable.

Definition 2.1.3.

Let $f : (X, \mathcal{M}) \rightarrow (Y, \mathcal{B})$ measurable. Define

$$\Sigma = \{E : E \subset Y, f^{-1}(E) \in \mathcal{M}\} \quad (2.1)$$

Proposition 2.1.4.

Σ is closed under countable unions, and complements making it a σ -algebra.

Theorem 2.1.5.

Let $f : (X, \mathcal{M}) \rightarrow \bar{\mathbb{R}}$. The following are equivalent:

- (i) f is measurable.
- (ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (v) $\{f \leq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

Theorem 2.1.6.

A function $f : X \rightarrow \bar{\mathbb{R}}$ is measurable if and only if

- (i) $f^{-1}(\{-\infty, \infty\}) \in \mathcal{M}$
- (ii) $f^{-1}((a, b)) \in \mathcal{M}$ for all open intervals $(a, b) \in \mathbb{R}$.

Lemma 2.1.7.

If f and g are measurable functions, then the following sets are measurable:

- (a) $X \cap \{x : f(x) > g(x)\}$
- (b) $X \cap \{x : f(x) \geq g(x)\}$
- (c) $X \cap \{x : f(x) = g(x)\}$

Theorem 2.1.8.

Let $f, g : X \rightarrow \bar{\mathbb{R}}$ be measurable functions, then $f + g$ and fg are measurable.

Example 2.1: The Cantor Lebesgue Function

We start the with regular cantor set $C = \bigcap_{k=1}^{\infty} E_k$ with each E_k having 2^k intervals of size $\frac{1}{3^k}$. Let

$$G_k = [0, 1] \setminus E_k \quad (2.2)$$

G_k has $2^k - 1$ intervals, $\{I_j^k\}_{j=1}^{2^k-1}$. Define $f_k : [0, 1] \rightarrow \mathbb{R}$ such that $f_k(0) = 0$, $f_k(1) = 1$ and $f_k(x) = \frac{j}{2^k}$ if $x \in I_j^k$. Let $x \in [0, 1]$. Then

$$|f_k(x) - f_{k+1}(x)| < \frac{1}{2^k} \quad (2.3)$$

and for every $m > 0$ we have

$$|f_k(x) - f_{k+m}(x)| \leq |f_k(x) - f_{k+1}(x)| + |f_{k+1}(x) - f_{k+2}(x)| + \cdots + |f_{k+m-1}(x) - f_{k+m}(x)| \quad (2.4)$$

$$< \frac{1}{2^k} + \frac{1}{2^{k+1}} + \cdots + \frac{1}{2^{k+m-1}} \quad (2.5)$$

$$< \sum_{i=k}^{\infty} \frac{1}{2^i} \quad (2.6)$$

$$= \frac{1}{2^{k-1}} \quad (2.7)$$

Then, for any $\varepsilon > 0$ there is some N such that $\frac{1}{2^{n-1}} < \varepsilon$, meaning $\{f_k\}$ is uniformly Cauchy in $[0, 1]$. This means that there is some limit function $f : [0, 1] \rightarrow \mathbb{R}$, and continuity of f_k means that f is then continuous.

We call this f the **Cantor-Lebesgue function**.

Proposition 2.1.9.

The Cantor-Lebesgue Function f has the following properties:

- (i) f is constant on each interval in $[0, 1] \setminus C$.
- (ii) f is nondecreasing.
- (iii) $f([0, 1]) = [0, 1]$.
- (iv) $f(C) = [0, 1]$.

Note that while Theorem 2.1.8 tells us that the product and sum of measurable functions are also measurable, the same is not true in general for the composition of Lebesgue measurable functions. We need the following assumptions to make this true:

Theorem 2.1.10.

Suppose $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is Lebesgue measurable and $g : \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}}$ is Borel measurable. Then $g \circ f$ is Lebesgue measurable.

Proof. If B is a Borel set in $\overline{\mathbb{R}}$ then

$$(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)) \quad (2.8)$$

We know that g is Borel measurable, which then implies that $g^{-1}(B)$ is Borel. Since f is Lebesgue measurable, then $f^{-1}(g^{-1}(B))$ is Lebesgue measurable. From here, we can conclude that $g \circ f$ is Lebesgue-measurable. $\square$

Corollary 2.1.11.

Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ be Lebesgue measurable.

- (i) Let $\varphi(x) = |f(x)|^p$ for $0 < p < \infty$, where φ assumes arbitrary extended values on the sets $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is Lebesgue measurable.
- (ii) Let $\varphi(x) = \frac{1}{f(x)}$, and let φ assume arbitrary extended values on the sets $f^{-1}(0)$, $f^{-1}(\infty)$, and $f^{-1}(-\infty)$. Then φ is Lebesgue measurable.

Definition 2.1.12: μ -Null Set.

Let $(X, \mathcal{M}, \mu)$ be a measure space. A measurable set N is called a **μ -null set** if $\mu(N) = 0$.

A property that holds for all $x \in X$ except for x in some μ -null set is said to hold **μ -almost everywhere**.

Theorem 2.1.13.

Let $(X, \mathcal{M}, \mu)$ be a complete measure space and let $f, g : X \rightarrow \overline{\mathbb{R}}$. If f is measurable and $f = g$ almost everywhere, then g is measurable.

2.2 Limits of Measurable Functions

For the remainder of this section, suppose all functions are $\overline{\mathbb{R}}$ valued.

Definition 2.2.1: Upper and Lower Envelopes.

Suppose $(X, \mathcal{M}, \mu)$ is a measure space, and $\{f_i\}$ is measurable on X . We define the **upper envelope** of $\{f_i\}$ by

$$\sup_i f_i(x) = \sup\{f_i(x) : i \in \mathbb{N}\} \quad (2.9)$$

and the **lower envelope** by

$$\inf_i f_i(x) = \inf\{f_i(x) : i \in \mathbb{N}\} \quad (2.10)$$

We also have definitions of $\limsup$ and $\liminf$ by

$$\limsup_{i \rightarrow \infty} f_i(x) = \inf_{j \geq 1} \left(\sup_{i \geq j} f_i(x) \right) \quad (2.11)$$

$$\liminf_{i \rightarrow \infty} f_i(x) = \sup_{j \geq 1} \left(\inf_{i \geq j} f_i(x) \right) \quad (2.12)$$

Theorem 2.2.2.

The functions

$$\sup_i f_i, \quad \inf_i f_i, \quad \limsup_{i \rightarrow \infty} f_i, \quad \liminf_{i \rightarrow \infty} f_i \quad (2.13)$$

are all measurable.

Definition 2.2.3.

Let $\{f_i\}$ be a sequence of measurable functions, such that

$$\lim_{i \rightarrow \infty} f_i(x) = f(x) \quad (2.14)$$

for μ -almost every $x \in X$. Then we say that f_i converges pointwise almost everywhere to f .

Theorem 2.2.4: Egoroff.

Let $(X, \mathcal{M}, \mu)$ be a finite measure space, and $\{f_j\}$ and $\{f\}$ be measurable and finite almost everywhere on X . Suppose $f_j \rightarrow f$ pointwise almost everywhere.

Then for every $\varepsilon > 0$, there exists some $A \in \mathcal{M}$ such that $\mu(A^c) < \varepsilon$ and $f_j \rightarrow f$ uniformly on A .

Corollary 2.2.5.

If you additionally assume that X is a metric space and μ is a Borel measure with $\mu(X) < \infty$, then A is a closed set.

Theorem 2.2.6.

Let $(X, \mathcal{M}, \mu)$ be a measure space with X a metric space, $\mathcal{M}$ a Borel σ -algebra, and $\mu(X) < \infty$. Let $\varepsilon > 0$.

Then for every $B \in \mathcal{M}$ there exists an open set U with $B \subset U$ such that $\mu(U \setminus B) < \varepsilon$ and a closed set F with $F \subset B$ with $\mu(B \setminus F) < \varepsilon$.

Definition 2.2.7: Almost Uniform Convergence.

We say that $f_i \rightarrow f$ **almost uniformly** if for all $\varepsilon > 0$ there exists some $A \in \mathcal{M}$ such that $\mu(A^c) < \varepsilon$ and $f_i \rightarrow f$ uniformly on A .

Example 2.2

Define $f_i = \chi_{[i, \infty]}$ on a measure space $(\mathbb{R}, \mathcal{M}, \lambda)$. Here, $f_i \rightarrow 0$ pointwise. However, $f_i \not\rightarrow 0$ uniformly on any set A such that $\mu(A^c) < \infty$. So then we say that $f_i \rightarrow 0$ almost uniformly.

Definition 2.2.8: Convergence in Measure.

Let $(X, \mathcal{M}, \mu)$ and $\{f_i\}$ be measurable functions on X . We say that $f_i \rightarrow f$ **in measure** if for any $\varepsilon > 0$

$$\lim_{i \rightarrow \infty} \mu(X \cap \{x : |f_i(x) - f(x)| \geq \varepsilon\}) = 0 \quad (2.15)$$

Theorem 2.2.9: Pointwise a.e. implies Convergence in Measure.

Let $(X, \mathcal{M}, \mu)$ be a finite measure space, $\{f_i\}$ and f measurable functions that are finite almost everywhere on X . If $f_i \rightarrow f$ pointwise almost everywhere on X , then $f_i \rightarrow f$ in measure.

Theorem 2.2.10.

Let $(X, \mathcal{M}, \mu)$ be a measure space, with $\{f_i\}$, f measurable functions such that $f_i \rightarrow f$ in measure. Then there exists a subsequence $\{f_{i_k}\}$ such that

$$\lim_{k \rightarrow \infty} f_{i_k}(x) = f(x) \quad (2.16)$$

for μ almost everywhere $x \in X$.

2.3 Approximations of Measurable Functions

We begin by definition simple functions, which are linear combinations of characteristic functions. For $\bigcup_{i=1}^k A_i = X$, then a simple function f is defined by

$$f(x) = \sum_{i=1}^k \alpha_i \chi_{A_i}(x) \quad (2.17)$$

for some constants α_i . Note that f has a finite, discrete range given by

$$\text{im}(f) = \{\alpha_i\}_{i=1}^k \quad (2.18)$$

Theorem 2.3.1.

Let $f : X \rightarrow \mathbb{R}$ be an arbitrary function. Then

- (i) There exists some sequence of simple functions $\{f_i\}$ such that $f_i(x) \rightarrow f(x)$ for all $x \in X$.
- (ii) If $f \geq 0$, $\{f_i\}$ can be chosen such that $f_i \uparrow f$.
- (iii) $\{f_i\}$ can be chosen such that $f_i \rightarrow f$ if and only if f is bounded.
- (iv) If f is measurable, then $\{f_i\}$ can be chosen to be measurable.

Theorem 2.3.2: Lusin.

Suppose $(X, \mathcal{M}, \mu)$ is a measure space where X is a metric space and μ is a finite Borel measure. Let $f : X \rightarrow \mathbb{R}$ be a measurable function that is finite almost everywhere.

Then for any $\varepsilon > 0$ there exists some closed $F \subset X$ with $\mu(F^c) < \varepsilon$ such that f is continuous on F in the relative topology.

3 Theory of Integration**3.1 Definition of the Integral****Definition 3.1.1.**

A function $f : X \rightarrow \mathbb{R}$ is called **countably-simple** if it assumes only a countable number of values, which could include $\pm\infty$.

Definition 3.1.2.

Let $(X, \mathcal{M}, \mu)$ be a measure space and a countably-simple measurable function $f : X \rightarrow \mathbb{R}$ with $f \geq 0$. Then

$$\int_X f d\mu = \sum_{i=1}^{\infty} a_i \mu(f^{-1}\{a_i\}) \quad (3.1)$$

Recall that we can always decompose $f = f^+ - f^-$ where

$$f^+ = f\chi_{\{x:f(x)\geq 0\}} \quad (3.2)$$

$$f^- = f\chi_{\{x:f(x)<0\}} \quad (3.3)$$

and that $|f| = f^+ + f^-$.

Definition 3.1.3.

If $\int_X f^+ d\mu < \infty$ or $\int_X f^- d\mu < \infty$, then

$$\int_X f d\mu := \int_X f^+ d\mu - \int_X f^- d\mu \quad (3.4)$$

Definition 3.1.4: Upper and Lower Integral.

Let $f : X \rightarrow \overline{\mathbb{R}}$. We define the **upper integral** of f by

$$\overline{\int}_X f d\mu := \inf \left\{ \int_X g d\mu : g \text{ measurable, countably-simple, } g \geq f \text{ } \mu\text{-a.e.} \right\} \quad (3.5)$$

and the **lower integral** of f by

$$\underline{\int}_X f d\mu := \sup \left\{ \int_X g d\mu : g \text{ measurable, countably-simple, } g \leq f \text{ } \mu\text{-a.e.} \right\} \quad (3.6)$$

Definition 3.1.5: Integrability.

Let $f : X \rightarrow \overline{\mathbb{R}}$ be measurable. If the upper and lower integrals coincide, then we write

$$\int_X f d\mu := \overline{\int}_X f d\mu = \underline{\int}_X f d\mu \quad (3.7)$$

If this integral is finite, then we say that f is **integrable**.

Theorem 3.1.6.

Let $f, g : X \rightarrow \overline{\mathbb{R}}$.

(i) If f is integrable, then f is finite μ -a.e.

(ii) If f, g are integrable, then for any $a, b \in \mathbb{R}$, $af + bg$ is integrable and

$$\int_X (af + bg) d\mu = a \int_X f d\mu + b \int_X g d\mu \quad (3.8)$$

(iii) If f, g are integrable and $f \leq g$ μ -a.e. then

$$\int_X f d\mu \leq \int_X g d\mu \quad (3.9)$$

(iv) If f is integrable and $E \in \mathcal{M}$, then $f\chi_E$ is integrable.

(v) f is integrable if and only if $|f|$ is integrable.

(vi) If f is integrable, then

$$\left| \int_X f d\mu \right| \leq \int_X |f| d\mu \quad (3.10)$$

Theorem 3.1.7.

If f is μ -measurable and $f \geq 0$ μ -a.e., then the integral of f exists.

Theorem 3.1.8.

Let $f \geq 0$ be a measurable function and g be integrable. Then

$$\int_X (f + g) d\mu = \int_X f d\mu + \int_X g d\mu \quad (3.11)$$

Corollary 3.1.9.

If f is measurable and f^+ is integrable or f^- is integrable, then f is integrable.

Theorem 3.1.10.

Let f be μ -measurable and g integrable such that $|f| \leq |g|$ μ almost everywhere. Then f is integrable.

3.2 Limits of Integrals

Lemma 3.2.1: Fatou's Lemma.

If $\{f_k\}$ is a sequence of nonnegative μ -measurable functions, then

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu \quad (3.12)$$

Theorem 3.2.2: Monotone Convergence Theorem.

If $\{f_k\}$ is a sequence of nonnegative μ -measurable functions such that $f_k \leq f_{k+1}$, then

$$\lim_{k \rightarrow \infty} \int_X f_k d\mu = \int_X \lim_{k \rightarrow \infty} f_k d\mu \quad (3.13)$$

Theorem 3.2.3.

Let $f_k \geq 0$ be μ -measurable. Then

$$\int_X \sum_{k=1}^{\infty} f_k d\mu = \sum_{k=1}^{\infty} \int_X f_k d\mu \quad (3.14)$$

Theorem 3.2.4.

If f is integrable and $\{E_k\}$ is a sequence of disjoint measurable sets such that

$$X = \bigcup_{k=1}^{\infty} E_k \quad (3.15)$$

then

$$\int_X f d\mu = \sum_{k=1}^{\infty} \int_{E_k} f d\mu \quad (3.16)$$

Corollary 3.2.5.

If $f \geq 0$ is integrable on a measure space $(X, \mathcal{M}, \mu)$, then if we define

$$\nu(E) = \int_E f d\mu \quad (3.17)$$

ν defines a measure on X

This corollary gives us a 1-1 correspondence between measurable functions and measures on a measure space.

Theorem 3.2.6: Dominated Convergence Theorem.

Suppose g is integrable, f is measurable, and $\{f_k\}$ is sequence such that $f_k \leq g$ for all k and $f_k \rightarrow f$ almost everywhere. Then

$$\lim_{k \rightarrow \infty} \int_X |f_k - f| d\mu = 0 \quad (3.18)$$

DCT essentially gives us criteria in which we can freely exchange limits and integrals, which is very powerful. Many proofs can be completed simply by stating "Invoke DCT. QED".

3.3 Relation to Riemann Integrals

We can further relate the Riemann and Lebesgue integral with the following:

Theorem 3.3.1.

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded Riemann integrable function then f is Lebesgue integrable and

$$\int_a^b f(x) dx = \int_{[a, b]} f d\lambda \quad (3.19)$$

Theorem 3.3.2.

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if f is continuous λ almost everywhere on $[a, b]$.

Definition 3.3.3: Improper Integral.

For $a \in \mathbb{R}$, let $f : [a, \infty) \rightarrow \mathbb{R}$ be a function that is Riemann integrable on each subinterval of $[a, \infty)$. We define the *improper integral* of f by

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx \quad (3.20)$$

If the improper integral is finite, then we say that the improper integral of f exists.

Theorem 3.3.4.

Let $f : [a, \infty) \rightarrow \mathbb{R}$ be a nonnegative function that is Riemann integrable on any subinterval of $[a, b]$. Then

$$\int_a^\infty f d\lambda = \lim_{b \rightarrow \infty} \int_a^b f dx \quad (3.21)$$

and f is then Lebesgue integral on $[a, \infty)$ if and only if the improper integral exists.

3.4 L^p Spaces**Definition 3.4.1: L^p Space.**

Let $1 \leq p \leq \infty$. Then $L^p(E, \mathcal{M}, \mu)$ denotes the class of all measurable functions such that

$$\|f\|_{L^p} < \infty \quad (3.22)$$

where

$$\|f\|_{L^p} = \begin{cases} \left(\int_E |f|^p \mu \right)^{1/p} & 1 \leq p < \infty \\ \inf \{M : |f| \leq M \mu\text{-a.e. on } E\} & p = \infty \end{cases} \quad (3.23)$$

Lemma 3.4.2.

L^p is a vector space. Furthermore, let $1 \leq p \leq \infty$. Then

(i) If $f, g \in L^p$, then $f + g \in L^p$.

(ii) If $f \in L^p$ and $c \in \mathbb{R}$, then $cf \in L^p$.

The following Lemma follows by basic arithmetic, but is useful for coming theorems:

Lemma 3.4.3.

Let $1 < p < \infty$, $a, b \geq 0$ and

$$\frac{1}{p} + \frac{1}{p'} = 1 \quad (3.24)$$

Then

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'} \quad (3.25)$$

The following theorem may be the most used theorem in all of PDE:

Theorem 3.4.4: Hölder's Inequality.

Let $1 \leq p \leq \infty$ and f, g measurable. Then

$$\int_X |fg| d\mu \leq \|f\|_{L^p} \|g\|_{L^{p'}} \quad (3.26)$$

Equality holds if and only if

$$\|f\|_p^p |g|^{p'} = \|g\|_{p'}^{p'} |f|^p \quad (3.27)$$

almost everywhere.

Definition 3.4.5.

We say that $1 \leq p, q \leq \infty$ are **Hölder conjugates** if

$$\frac{1}{p} + \frac{1}{q} = 1 \quad (3.28)$$

The most common conjugates are 2, 2 and $1, \infty$.

Theorem 3.4.6: L^p Duals.

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space, f measurable with $1 \leq p \leq \infty$ and p, q are Hölder conjugates. Then

$$\|f\|_p = \sup \left\{ \int_X fg \, d\mu : \|g\|_q \leq 1 \right\} \quad (3.29)$$

Theorem 3.4.7: Minkowski's Inequality.

Let $1 \leq p \leq \infty$ and $f, g \in L^p$. Then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p \quad (3.30)$$

We can also say something about the inclusions of L^p spaces with the following:

Theorem 3.4.8.

Let $\mu(X) < \infty$ and $1 \leq p \leq q \leq \infty$. Then $L^q(X) \subseteq L^p(X)$.

Theorem 3.4.9.

If $1 \leq p \leq \infty$ then $L^p(X)$ is a complete metric space.

Lemma 3.4.10.

Let f be an integrable function. Then for every $\varepsilon > 0$, there exists some $\delta > 0$ such that for all $A \subset X$ with $\mu(A) < \delta$ there is $\int_A |f| \, d\mu < \varepsilon$.

Corollary 3.4.11.

Let $f \in L^p(X)$ with $1 \leq p \leq \infty$. Then for all $\varepsilon > 0$ there is some $\delta > 0$ such that for all $A \subset X$ with $\mu(A) < \delta$, then $\int_A |f|^p \, d\mu < \varepsilon$.

Lemma 3.4.12.

Let f be an integrable function. Then for all $\varepsilon > 0$ there is some $E \in \mathcal{M}$ with $\mu(E) < \infty$ such that $\int_{E^c} |f| \, d\mu < \varepsilon$.

Corollary 3.4.13.

Let f be an integrable function. Then for all $\varepsilon > 0$ there is some $E \in \mathcal{M}$ with $\mu(E) < \infty$ such that $\int_{E^c} |f|^p \, d\mu < \varepsilon$.

Theorem 3.4.14: Vitali's Convergence Theorem.

Suppose $\{f_k\}, f \in L^p(X)$ with $1 \leq p < \infty$. Then $\|f_k - f\|_p \rightarrow 0$ if the following hold:

- (i) $f_k \rightarrow f$ μ -almost everywhere.
- (ii) For all $\varepsilon > 0$ there exists some $E \in \mathcal{M}$ such that $\mu(E) < \infty$ and $\int_{E^c} |f_k|^p \, d\mu < \varepsilon$ for all $k \in \mathbb{N}$.
- (iii) For all $\varepsilon > 0$ there exists some $\delta > 0$ such that for all $\mu(E) < \delta$ we have $\int_{E^c} |f_k|^p \, d\mu < \varepsilon$ for all $k \in \mathbb{N}$.

Definition 3.4.15: Linear Functional.

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then $F : L^p(X) \rightarrow \mathbb{R}$ is a linear functional on $L^p(X)$ if

$$F(af + bg) = aF(f) + bF(g) \quad (3.31)$$

Definition 3.4.16: Bounded Functional.

A linear functional F is bounded if

$$\|F\| := \sup \{ |F(f)| : f \in L^p(X), \|f\|_p \leq 1 \} < \infty \quad (3.32)$$

Theorem 3.4.17.

A linear functional F is bounded if and only if it is continuous.

Theorem 3.4.18.

Let $\mathcal{B}(L^p(X), \mathbb{R})$ denote the set of all bounded linear mappings of $L^p(X) \rightarrow \mathbb{R}$. $\mathcal{B}$ is a Banach space with respect to the norm defined in (3.32).

Theorem 3.4.19.

If $1 \leq p \leq \infty$ and p, q are Hölder conjugates and $g \in L^q(X)$ then

$$F(f) = \int_X fg \, d\mu \quad (3.33)$$

defines a bounded linear functional on $L^p(X)$ with $\|F\| = \|g\|_q$

We can also uniquely define a function which creates a linear functional on the dual space with the following theorem:

Theorem 3.4.20.

If $1 < p < \infty$ and F is a bounded linear functional on $L^p(X)$, then there exists some $g \in L^q(X)$ such that for all $f \in L^p(X)$

$$F(f) = \int_X fg \, d\mu \quad (3.34)$$

We also have that g is unique with $\|g\|_q = \|F\|$.

Note that this still holds for $p = 1$, we just need to further assume that μ is σ -finite.

3.5 Signed Measures

Our previous definition of measure only allowed them to take positive real values. We can extend our definition to take negative values by the following:

Definition 3.5.1: Signed Measure.

An extended real valued function ν defined on a σ -algebra $\mathcal{M}$ is a **signed measure** if

- (i) ν assumes at most one of $\infty, -\infty$.
- (ii) $\nu(\emptyset) = 0$
- (iii) If $\{E_k\}$ is a disjoint sequence of measurable sets then

$$\nu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \nu(E_k) \quad (3.35)$$

where the series either converges absolutely or diverges to $\pm\infty$.

Definition 3.5.2: Absolute Continuity of Signed Measures.

Let $(X, \mathcal{M}, \mu)$ be a measure space and ν a signed measure defined on $\mathcal{M}$. We say that ν is absolutely continuous with respect to μ , $\nu \ll \mu$, if $\mu(E) = 0$ implies that $\nu(E) = 0$.

Theorem 3.5.3: Hahn Decomposition.

Let $X = P \cup N$ with $P \cap N = \emptyset$. Then a signed measure ν has $\nu(E) \geq 0$, for all $E \subset P$ and $\nu(E) \leq 0$ for all $E \subset N$. We call P the **positive set** and N the **negative set**.

Note that the Hahn decomposition is not unique if ν has a non-empty null set.

Definition 3.5.4: Mutually Singular Measures.

Let μ_1, μ_2 be measures in $(X, \mathcal{M})$. We say that μ_1, μ_2 are **mutually singular**, $\mu_1 \perp \mu_2$ if there exists some $E \in \mathcal{M}$ such that $\mu_1(E) = \mu_2(X \setminus E) = 0$.

Theorem 3.5.5: Jordan Decomposition.

Let ν be a signed measure on $\mathcal{M}$. Then there exists measures ν^+, ν^- where at least one is finite such that for every $E \in \mathcal{M}$ we have

$$\nu(E) = \nu^+ - \nu^-(E) \quad (3.36)$$

Theorem 3.5.6: Radon-Nikodym.

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space and ν a σ -finite signed measure on $\mathcal{M}$ such that $\nu \ll \mu$. Then there exists a measurable f such that f^+ or f^- is integrable and for every $E \in \mathcal{M}$

$$\nu(E) = \int_E f \, d\mu \quad (3.37)$$

Note that $f := \frac{d\nu}{d\mu}$ is called the **Radon-Nikodym derivative** of ν with respect to μ .

3.6 Product Measures

Let μ and ν be measures on X and Y defined on σ algebras $\mathcal{M}_X$ and $\mathcal{M}_Y$ respectively. We wish to define a measure in $X \times Y$.

Definition 3.6.1.

For each $S \subset X \times Y$ define

$$\varphi(S) = \inf \left\{ \sum_{j=1}^{\infty} \mu(A_j) \nu(B_j) \right\} \quad (3.38)$$

where the infimum is taken over all sequences $\{A_j \times B_j\}$ such that $S \subset \bigcup_{j=1}^{\infty} A_j \times B_j$ for $A_j \in \mathcal{M}_X$ and $B_j \in \mathcal{M}_Y$.

Proposition 3.6.2.

φ is an outer measure on $X \times Y$.

Theorem 3.6.3: Fubini.

Suppose $(X, \mathcal{M}_X, \mu)$ and $(Y, \mathcal{M}_Y, \nu)$ are complete measure spaces.

(i) If $A \in \mathcal{M}_X$ and $B \in \mathcal{M}_Y$, then $A \times B \in \mathcal{M}_{X \times Y}$ and

$$(\mu \times \nu)(A \times B) = \mu(A) \nu(B) \quad (3.39)$$

(ii) If $S \in \mathcal{M}_{X \times Y}$ and S is σ -finite with respect to $\mu \times \nu$ then

$$S_y = \{x : (x, y) \in S\} \in \mathcal{M}_X \quad \nu \text{ a.e. } y \in Y \quad (3.40)$$

$$S_x = \{y : (x, y) \in S\} \in \mathcal{M}_Y \quad \mu \text{ a.e. } x \in X \quad (3.41)$$

$y \mapsto \mu(S_y)$ is $\mathcal{M}_Y$ measurable and $x \mapsto \nu(S_x)$ is $\mathcal{M}_X$ measurable and

$$(\mu \times \nu)(S) = \int_X \nu(S_x) d\mu(x) = \int_Y \mu(S_y) d\nu(y) \quad (3.42)$$

(iii) If $f \in L^1(X \times Y, \mathcal{M}_{X \times Y}, \mu \times \nu)$ then $y \mapsto f(x, y)$ is ν integrable for μ -a.e. $x \in X$ and $x \mapsto f(x, y)$ is μ integrable for ν -a.e. $y \in Y$. Furthermore, $x \mapsto \int_Y f(x, y) d\nu(y)$ is μ -integrable and $y \mapsto \int_X f(x, y) d\mu(x)$ is ν -integrable and

$$\int_{X \times Y} f d(\mu \times \nu) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y) \quad (3.43)$$

Corollary 3.6.4: Tonelli.

Let $f \geq 0$ be $\mathcal{M}_{X \times Y}$ -measurable and $\{(x, y) : f(x, y) \neq 0\}$ is σ -finite with respect to $\mu \times \nu$. Then then $y \mapsto f(x, y)$ is ν integrable for μ -a.e. $x \in X$ and $x \mapsto f(x, y)$ is μ integrable for ν -a.e. $y \in Y$. Furthermore, $x \mapsto \int_Y f(x, y) d\nu(y)$ is μ -integrable and $y \mapsto \int_X f(x, y) d\mu(x)$ is ν -integrable and

$$\int_{X \times Y} f d(\mu \times \nu) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y) \quad (3.44)$$

Definition 3.6.5: Convolution.

Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be Lebesgue measurable functions. The convolution is defined as

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y) g(x - y) dy \quad (3.45)$$

Lemma 3.6.6.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be Lebesgue measurable. Define $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$F(x, y) = f(x - y) \quad (3.46)$$

Then F is λ_{2n} -measurable.

Theorem 3.6.7: Young's Convolution Inequality.

If $f \in L^p(\mathbb{R}^n)$ with $1 \leq p \leq \infty$ and $g \in L^1(\mathbb{R}^n)$ then $f * g \in L^p(\mathbb{R}^n)$ and

$$\|f * g\|_p \leq \|f\|_p \|g\|_1 \quad (3.47)$$

Proposition 3.6.8.

Fix some $g \in L^1(\mathbb{R}^n)$. Let $T : L^p(\mathbb{R}^n) \rightarrow L^p(\mathbb{R}^n)$ be defined by $T(f) = f * g$. Then T is a bounded linear map.

3.7 Distribution Functions

Note: These are not the same as generalized functions which are also called distributions.

Definition 3.7.1.

Let $(X, \mathcal{M}, \mu)$ be a complete σ -finite measure space and f a measurable function on X . Let $t \in \mathbb{R}$. Define

$$E_t = \{x : f(x) > t\} \in \mathcal{M} \quad (3.48)$$

Let $A_f(t) = \mu(E_t)$. The map $t \mapsto A_f(t)$ is called a **distribution function**.

Theorem 3.7.2.

Let $f \geq 0$ be measurable. Then

$$\int_X f \, d\mu = \int_{[0, \infty)} A_f \, d\lambda \quad (3.49)$$

Theorem 3.7.3.

If f is measurable and $1 \leq p < \infty$ then

$$\int_X |f|^p \, d\mu = p \int_{[0, \infty)} t^{p-1} \mu(\{x : |f(x)| > t\}) \, d\lambda(t) \quad (3.50)$$

Theorem 3.7.4: Vitali's Covering Theorem.

Let $\mathcal{F}$ be a collection of non-degenerate closed balls in $\mathbb{R}^n$ with

$$\sup\{\text{diam } D : D \in \mathcal{F}\} < \infty \quad (3.51)$$

Then there exists a countable family $\mathcal{G}$ of disjoint balls in $\mathcal{F}$ such that

$$\bigcup_{B \in \mathcal{F}} B \subset \bigcup_{\hat{B} \in \mathcal{G}} \hat{B} \quad (3.52)$$

Corollary 3.7.5.

Let $U \subset \mathbb{R}^n$ be open and $\delta > 0$. Then there exists a countable collection $\mathcal{S}$ of disjoint closed balls in U such that $\text{diam } B \leq \delta$ for all $B \in \mathcal{S}$ and

$$\lambda_n \left(U \setminus \bigcup_{B \in \mathcal{S}} B \right) = 0 \quad (3.53)$$

3.8 Lebesgue Points

Definition 3.8.1: Maximal Function.

Let $f \in L^1(\mathbb{R}^n)$. We define Mf , the maximal function by

$$Mf(x) = \sup_{r>0} \text{P.V.} \int_{B(x,r)} |f| \, d\lambda \quad (3.54)$$

where the Cauchy Principal Value is defined by

$$\text{P.V.} \int_E |f| \, d\lambda = \frac{1}{\lambda(E)} \int_E |f| \, d\lambda \quad (3.55)$$

Proposition 3.8.2.

$Mf : \mathbb{R}^n \rightarrow \mathbb{R}$ is a non-negative function which is measurable and not in $L^1(\mathbb{R}^n)$.

Theorem 3.8.3: Hardy-Littlewood.

Let $f \in L^1(\mathbb{R}^n)$. Then for all $t > 0$

$$\lambda(\{Mf > t\}) \leq \frac{5^n}{t} \int_{\mathbb{R}^n} |f| \, d\lambda \quad (3.56)$$

Theorem 3.8.4.

If $f \in L^1_{loc}(\mathbb{R}^n)$ then

$$\lim_{r \rightarrow 0} \text{P.V.} \int_{B(x,r)} f(y) d\lambda(y) = f(x) \quad (3.57)$$

almost everywhere $x \in \mathbb{R}^n$.

Theorem 3.8.5.

If $f \in L^1_{loc}(\mathbb{R}^n)$ then

$$\lim_{r \rightarrow 0} \text{P.V.} \int_{B(x,r)} |f(y) - f(x)| d\lambda(y) = 0 \quad (3.58)$$

3.9 Radon-Nikodym Derivatives

For the remainder of this section, all measures are defined on $\mathbb{R}^n$.

Definition 3.9.1: Radon Measure.

A Borel measure on $\mathbb{R}^n$ that is finite on compact sets is called a **Radon measure**.

Example 3.1

The Lebesgue measure λ is a Radon measure, but the Hausdorff measure H^s for $0 < s < n$ is not since

$$H^2([0, 1]^3) = \infty \quad (3.59)$$

Lemma 3.9.2.

If $\mu \ll \lambda$ for a Radon measure μ , then

$$f(x) = D_\lambda \mu(x) = \lim_{r \rightarrow 0} \frac{\mu(B(x, r))}{\lambda(B(x, r))} \quad (3.60)$$

λ almost everywhere.

Theorem 3.9.3.

Let μ be a Radon measure that is singular with respect to λ , that is μ is concentrated on a Borel set A and $\mu(A^c) = \lambda(A) = 0$, then

$$D_\lambda \mu(x) := \lim_{r \rightarrow 0} \frac{\mu(B(x, r))}{\lambda(B(x, r))} = 0 \quad (3.61)$$

λ -almost everywhere x .

Theorem 3.9.4.

Let μ and ν be Radon measures on $\mathbb{R}^n$, and ν be Borel regular. Then

$$D_\mu \nu(x) = \lim_{r \rightarrow 0} \frac{\nu(B(x, r))}{\mu(B(x, r))} \quad (3.62)$$

exists and is finite for μ -a.e. x . Moreover $D_\mu \nu$ is μ -measurable.

Theorem 3.9.5: Lebesgue Decomposition for Radon Measures.

Suppose ν is a Radon measure on $\mathbb{R}^n$. Then there exists measures μ and σ such that:

$$\nu = \mu + \sigma, \quad \mu \ll \lambda, \quad \sigma \perp \lambda \quad (3.63)$$

Theorem 3.9.6.

Suppose ν is a Radon measure on $\mathbb{R}^n$. Let $\nu = \mu + \sigma$ be its Lebesgue decomposition. Let $f = D_\lambda \mu$. Then

$$\lim_{r \rightarrow 0} \frac{\nu(B(x, r))}{\lambda(B(x, r))} = f(x) \quad (3.64)$$

λ -almost everywhere $x \in \mathbb{R}^n$.

Definition 3.9.7: Regular Differentiation Basis.

Let $x \in \mathbb{R}^n$ and $\{E_k(x)\}$ be a sequence of Borel sets. We say that $\{E_k(x)\}$ is a **regular differentiation basis** at x if there exists $\alpha_x > 0$ such that:

There is a sequence of balls $B(x, r_k)$ with $r_k \rightarrow 0$ such that $E_k(x) \subset B(x, r_k)$ and $\lambda(E_k(x)) \geq \alpha_x \lambda(B(x, r_k))$.

Theorem 3.9.8.

Let ν be a Radon measure on $\mathbb{R}^n$. Consider its Lebesgue decomposition $\nu = \mu + \sigma$. Let $f = D_\lambda \mu$. Then for λ -a.e. $x \in \mathbb{R}^n$

$$\lim_{k \rightarrow \infty} \frac{\sigma(E_k(x))}{\lambda(E_k(x))} = 0 \quad \text{and} \quad \lim_{k \rightarrow \infty} \frac{\mu(E_k(x))}{\lambda(E_k(x))} = f(x) \quad (3.65)$$

for every $\{E_k(x)\}$ regular differentiation basis at x .

Theorem 3.9.9.

Let f be a Lebesgue integrable function defined on $[a, b]$. For each $x \in [a, b]$ let:

$$F(x) = \int_a^x f(t) d\lambda(t) \quad (3.66)$$

Then $F'(x) = f(x)$ for x λ -a.e.

3.10 Functions of Bounded Variation**Definition 3.10.1: Total Variation.**

Let $f : [a, b] \rightarrow \mathbb{R}$. The total variation of f from a to x , $x \leq b$ is defined by

$$V_f(a; x) = \sup \sum_{i=1}^k |f(t_i) - f(t_{i-1})| \quad (3.67)$$

where the supremum is taken over all finite sequences $a = t_0 < t_1 < \dots < t_k = x$. We say that f is a function of bounded variation on $[a, b]$; that is $f \in BV([a, b])$, if $V_f(a; b) < \infty$.

It is trivial to show that if a function is in $BV([a, b])$ then it is bounded on $[a, b]$.

Theorem 3.10.2.

Let $f \in BV([a, b])$. Then $f = f_1 - f_2$ where both f_1 and f_2 are non-decreasing.

Definition 3.10.3.

A function $f : [a, b] \rightarrow \mathbb{R}$ is **absolutely continuous** on $[a, b]$, written as $f \in AC([a, b])$, if for all $\varepsilon > 0$ if there exists some $\delta > 0$ such that

$$\sum_{i=1}^k |f(b_i) - f(a_i)| < \varepsilon \quad (3.68)$$

for any collection of non-overlapping intervals $[a_1, b_1], [a_2, b_2], \dots, [a_k, b_k]$ on $[a, b]$ with

$$\sum_{i=1}^k |b_i - a_i| < \delta \quad (3.69)$$

Definition 3.10.4: Condition N.

A function $f : [a, b] \rightarrow \mathbb{R}$ is said to satisfy **condition N** if $E \subset [a, b]$, $\lambda(E) = 0$ implies that $\lambda(f(E)) = 0$.

Proposition 3.10.5.

Lipschitz functions automatically satisfy Condition N.

Theorem 3.10.6.

If $f \in AC([a, b])$, then f satisfies Condition N

Theorem 3.10.7.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a non-decreasing function. Then $f'(x)$ exists λ -a.e. $x \in \mathbb{R}$.

Theorem 3.10.8.

If $f \in AC([a, b])$ then $f \in BV([a, b])$.

Theorem 3.10.9.

If $f \in AC([a, b])$ with the property that $f' \equiv 0$ almost everywhere, then f is constant.

Theorem 3.10.10.

Let $f \in BV([a, b])$. Then

$$\int_a^b |f'(x)| d\lambda(t) \leq V_f(b) \quad (3.70)$$

Theorem 3.10.11: Fundamental Theorem of Calculus.

The function $f : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if and only if f' exists almost everywhere, $f' \in L^1([a, b])$ and

$$f(x) - f(a) = \int_a^x f'(t) d\lambda(t) \quad (3.71)$$

for all $x \in [a, b]$.

Part II

Previous Qualifying Exams

These are problems from past Purdue qualifying exams which can be found <https://www.math.purdue.edu/academic/grad/qualexams.html>. Note that all solutions are either mine or sourced from other resources, so the validity of them should not be assumed. I have tried to reference relevant theorems in the notes section above, but some results may be assumed as true when they were proved in class or homework.

Exam 1: Fall 2013

Problem 1.1: 15 points

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f, f_1, f_2, \dots$ be non-negative integrable functions such that $f_n \rightarrow f$ almost everywhere and $\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu$. If $E \in \mathcal{M}$, then show that $\lim_{n \rightarrow \infty} \int_E f_n d\mu = \int_E f d\mu$.

Solution to Problem 1.1: Since $f_n \rightarrow f$ almost everywhere and each f_n is positive, Fatou's Lemma (Lemma 3.2.1) gives us that

$$\int_E f d\mu \leq \liminf_{n \rightarrow \infty} \int_E f_n d\mu \quad (1.1.1)$$

$$\int_{X \setminus E} f d\mu \leq \liminf_{n \rightarrow \infty} \int_{X \setminus E} f_n d\mu \quad (1.1.2)$$

We can then split $X = E \cup (X \setminus E)$ and see

$$\int_X f_n d\mu = \int_E f_n d\mu + \int_{X \setminus E} f_n d\mu \quad (1.1.3)$$

So then taking $n \rightarrow \infty$ we have

$$\int_X f d\mu = \lim_{n \rightarrow \infty} \int_E f_n d\mu + \lim_{n \rightarrow \infty} \int_{X \setminus E} f_n d\mu \quad (1.1.4)$$

From our application of Fatou, we also have

$$\int_X f d\mu = \int_E f d\mu + \int_{X \setminus E} f d\mu \leq \liminf_{n \rightarrow \infty} \int_E f_n d\mu + \liminf_{n \rightarrow \infty} \int_{X \setminus E} f_n d\mu \quad (1.1.5)$$

Since the limit is less than the liminf of the same object, then we have equality and

$$\int_E f d\mu + \int_{X \setminus E} f d\mu = \lim_{n \rightarrow \infty} \int_E f_n d\mu + \lim_{n \rightarrow \infty} \int_{X \setminus E} f_n d\mu \quad (1.1.6)$$

From which we can conclude $\int_E f d\mu = \lim_{n \rightarrow \infty} \int_E f_n d\mu$. $\square$

Problem 1.2: 10 points (a), 5 points (b)

- (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lebesgue integrable function such that $\int_a^b f(x) d\lambda(x) = 0$ for every $a < b$. Show that $f(x) = 0$ for almost every x .
- (b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a Lebesgue integrable function such that

$$\int_R f(x)g(x) d\lambda(x) = 0 \quad (1.2.1)$$

for every continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$. Show that $f(x) = 0$ for almost every x .

Solution to Problem 1.2:

- (a) We begin by decomposing $f = f^+ - f^-$ and some $E \subset \mathbb{R}$ such that $E = \bigcup_{k=1}^n (a_k, b_k)$ for a collection of disjoint intervals (a_k, b_k) . We can then apply the finite additivity of the integral to see

$$\int_E f d\lambda(x) = \sum_{k=1}^n \int_{a_k}^{b_k} f(x) d\lambda(x) = 0 \quad (1.2.2)$$

So then the integral of f over any measurable set is zero. Now, split E into $E_n^+ = \{x : f(x) > 1/n\}$ and $E_n^- = \{x : f(x) < -1/n\}$. We then have

$$0 = \int_E f d\lambda(x) = \int_{E_n^+} f^+ d\lambda(x) - \int_{E_n^-} f^- d\lambda(x) \quad (1.2.3)$$

$$\geq \int_{E_n^+} \frac{1}{n} d\lambda(x) + \int_{E_n^-} \frac{1}{n} d\lambda(x) \quad (1.2.4)$$

$$= \frac{\lambda(E_n^+) + \lambda(E_n^-)}{n} \quad (1.2.5)$$

So then the sum of the measures of each set which f is nonzero is 0, so $f(x) = 0$ almost everywhere.

- (b) Choose some sequence of continuous, compactly supported function g_n such that $g_n \rightarrow \chi_{(a,b)}$ pointwise for some arbitrary interval (a, b) . Since each g_n must be bounded, we can apply DCT (Theorem 3.2.6) to see

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(x)g_n(x) d\lambda(x) = \int_a^b f(x) d\lambda(x) = 0 \quad (1.2.6)$$

Then by part (a), we have that $f(x) = 0$ almost everywhere. □

Problem 1.3: 15 points

Show that $f(x) = \sin(x^2)$ is not Lebesgue integrable over $[0, \infty)$. However, show that the improper Riemann integral $\int_0^\infty \sin(x^2) dx$ exists.

Solution to Problem 1.3: We first show that $f(x)$ is not Lebesgue integrable. By the definition of integrability, we need

$$\int_0^\infty |\sin(x^2)| d\lambda(x) < \infty \quad (1.3.1)$$

Split $A = [0, \infty)$ into intervals $I_k = [\sqrt{k\pi}, \sqrt{(k+1)\pi}]$ for $k \in \mathbb{N}_0$. So then we have

$$\int_A |\sin(x^2)| d\lambda(x) = \sum_{k=0}^{\infty} \int_{I_k} |\sin(x^2)| d\lambda(x) \quad (1.3.2)$$

Since this is a sum, we can pick any singular I_k and say that

$$\int_A |\sin(x^2)| d\lambda(x) \geq \int_{I_k} |\sin(x^2)| d\lambda(x) \geq C\lambda(I_k) = C(\sqrt{(k+1)\pi} - \sqrt{k\pi}) \quad (1.3.3)$$

for some constant C . So then using this estimate, we have

$$\int_A |\sin(x^2)| d\lambda(x) \geq C \sum_{k=0}^{\infty} \sqrt{(k+1)\pi} - \sqrt{k\pi} = \infty \quad (1.3.4)$$

So then f is not Lebesgue integrable.

We now show that the Riemann integral exists. Back to calculus, wahoo! We see

$$\int_0^{\infty} \sin(x^2) dx = \lim_{R \rightarrow \infty} \int_0^R \sin(x^2) dx \quad (1.3.5)$$

We can proceed via integration by parts. We see

$$\int_0^R \sin(x^2) dx = \int_0^R \frac{1}{2x} 2x \sin(x^2) dx \quad (1.3.6)$$

$$= \left. \frac{-\cos(x^2)}{2x} \right|_0^R - \int_0^R \frac{\cos(x^2)}{2x^2} dx \quad (1.3.7)$$

The boundary terms die off as $R \rightarrow \infty$, and $\cos(x^2)/2x^2$ is bounded by $1/x^2$ so it is absolutely integrable. $\square$

Problem 1.4: 10 Points

Let $(X, \mathcal{M}, \mu)$ be a measure space, T be a metric space, and $f : X \times T \rightarrow \mathbb{R}$ a function. Assume that $f(\cdot, t)$ is a measurable function for each $t \in T$ and $f(x, \cdot)$ is a continuous function for each $x \in X$. Assume also that there exists an integrable function g such that for each $t \in T$ we have $|f(x, t)| \leq g(x)$ for almost all $x \in X$. Show that the function $F : T \rightarrow \mathbb{R}$ defined by

$$F(t) = \int_X f(x, t) d\mu(x) \quad (1.4.1)$$

is a continuous function.

Solution to Problem 1.4: Pick some sequence $t_n \rightarrow t$. Since $f(x, \cdot)$ is continuous for any x , we can assume that $f(x, t_n) \rightarrow f(x, t)$ for all $x \in X$. By assumption, each $f(x, t)$ is bounded by an L^1 function in g , so then we have

$$|f(x, t_n) - f(x, t)| \leq |f(x, t_n)| + |f(x, t)| \leq 2g(x) \quad (1.4.2)$$

So then by linearity of the integral, we have

$$|F(t_n) - F(t)| = \left| \int_X f(x, t_n) d\mu(x) - \int_X f(x, t) d\mu(x) \right| \quad (1.4.3)$$

$$= \left| \int_X f(x, t_n) - f(x, t) d\mu(x) \right| \quad (1.4.4)$$

$$\leq \int_X |f(x, t_n) - f(x, t)| d\mu(x) \quad (1.4.5)$$

Which tends to 0 as $n \rightarrow \infty$, so then we can conclude that $F(t_n) \rightarrow F(t)$ meaning it is continuous. $\square$

Problem 1.5: 15 Points

Prove the following version of Minkowski's inequality for $1 \leq p < \infty$. If $f \geq 0$ is a measurable function on $\mathbb{R}^2$, then

$$\left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) d\lambda(x) \right)^p d\lambda(y) \right)^{1/p} \leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)^p d\lambda(y) \right)^{1/p} d\lambda(x) \quad (1.5.1)$$

Solution to Problem 1.5: Let q be the Hölder conjugate of p . By Theorem 3.4.6, we can write

$$\left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) d\lambda(x) \right)^p d\lambda(y) \right)^{1/p} = \left\| \int_{\mathbb{R}} f(x, \cdot) d\lambda(x) \right\|_p \quad (1.5.2)$$

$$= \sup \left\{ \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) d\lambda(x) \right) g(y) d\lambda(y) : \|g\|_q \leq 1 \right\} \quad (1.5.3)$$

Without loss of generality, we can suppose that $g \geq 0$ and take $\|g\|_q = 1$ so the inequality still holds when removing the supremum. Since $f \geq 0$, we can then apply Tonelli (Corollary 3.6.4) to see

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) d\lambda(x) \right) g(y) d\lambda(y) = \int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) g(y) d\lambda(x) d\lambda(y) \quad (1.5.4)$$

We now wish to apply Hölder (Theorem 3.4.4), seeing

$$\int_{\mathbb{R}} \int_{\mathbb{R}} f(x, y) g(y) d\lambda(x) d\lambda(y) \leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)^p d\lambda(y) \right)^{1/p} \left(\int_{\mathbb{R}} g(y)^q d\lambda(y) \right)^{1/q} d\lambda(x) \quad (1.5.5)$$

Since we assumed $\|g\|_q = 1$ we then have

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)^p d\lambda(y) \right)^{1/p} \left(\int_{\mathbb{R}} g(y)^q d\lambda(y) \right)^{1/q} d\lambda(x) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)^p d\lambda(y) \right)^{1/p} d\lambda(x) \quad (1.5.6)$$

Combining all of our inequalities, we then have

$$\left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) d\lambda(x) \right)^p d\lambda(y) \right)^{1/p} \leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y)^p d\lambda(y) \right)^{1/p} d\lambda(x) \quad (1.5.7)$$

as desired. $\square$

Problem 1.6: 15 Points

Let $\{f_n\}_{n \geq 1}$ be a sequence of absolutely continuous real valued functions on $[0, 1]$ such that

(i) $f(x) = \sum_{n=1}^{\infty} f_n(x)$ converges for every $x \in [0, 1]$.

(ii) $\int_0^1 (\sum_{n=1}^{\infty} |f'_n(x)|) d\lambda(x) < \infty$.

Show that f is absolutely continuous on $[0, 1]$.

Solution to Problem 1.6: We know from (ii) that if $G(x) := \sum_{n=1}^{\infty} |f'_n(x)|$ then

$$\int_0^1 G(x) d\lambda(x) < \infty \quad (1.6.1)$$

By the Riemann FTC and Tonelli (Corollary 3.6.4) we can determine the derivative of f .

$$f(x) - f(0) = \sum_{n=1}^{\infty} (f_n(x) - f_n(0)) \quad (1.6.2)$$

$$= \sum_{n=1}^{\infty} \int_0^x f'_n(t) d\lambda(t) \quad (1.6.3)$$

$$= \int_0^x \sum_{n=1}^{\infty} f'_n(t) d\lambda(t) \quad (1.6.4)$$

So then we can see

$$\left| \int_0^x \sum_{n=1}^{\infty} f'_n(t) d\lambda(t) \right| \leq \int_0^x \sum_{n=1}^{\infty} |f'_n(t)| d\lambda(t) < \infty \quad (1.6.5)$$

So then by Theorem 3.10.11, we have that f must be absolutely continuous. $\square$

Problem 1.7: 15 Points

Let $f_n, f \in BV([0, 1])$ and $n \geq 1$. Assume that $\sum_{n=1}^{\infty} V_{f_n - f}(0; 1) < \infty$. Show that $f'_n \rightarrow f'$ λ -almost everywhere on $[0, 1]$.

Solution to Problem 1.7: Define the function $g_n = f - f_n$. Since BV is a normed Banach space, so $\sum_{n=1}^{\infty} g_n$ converges, particularly each partial sum. So then

$$G_N = \sum_{n=1}^N (f_n - f) = f_N - f \quad (1.7.1)$$

Since $f_n \rightarrow f$ pointwise, we have that the partial sums $G_N \equiv 0$ for all N . So then $\sum_{n=1}^{\infty} g_n = 0$. The derivatives of functions of bounded variations exist λ -almost everywhere by Theorem 3.10.10 and we have

$$\int_0^1 |g'_n(x)| d\lambda(x) \leq V_{g_n}(0; 1) \quad (1.7.2)$$

We can then apply Tonelli (Corollary 3.6.4) to see

$$\sum_{n=1}^{\infty} \int_0^1 |g'_n(x)| d\lambda(x) = \int_0^1 \sum_{n=1}^{\infty} |g'_n(x)| d\lambda(x) < \infty \quad (1.7.3)$$

Therefore we must have $\sum_{n=1}^{\infty} |g'_n(x)| < \infty$ which means that $g'_n \rightarrow 0$ and $f'_n \rightarrow f'$ almost everywhere. $\square$

References

- [1] Monica Torres. *Lecture Notes: Lebesgue Theory of Integration*. URL: <https://www.math.purdue.edu/~torresm/lebesgue-theory.html>.
- [2] William P. Ziemer and Monica Torres. *Modern Real Analysis*. Vol. 278. Graduate Texts in Mathematics. Cham, Switzerland: Springer, 2017. ISBN: 978-3-319-64628-2.